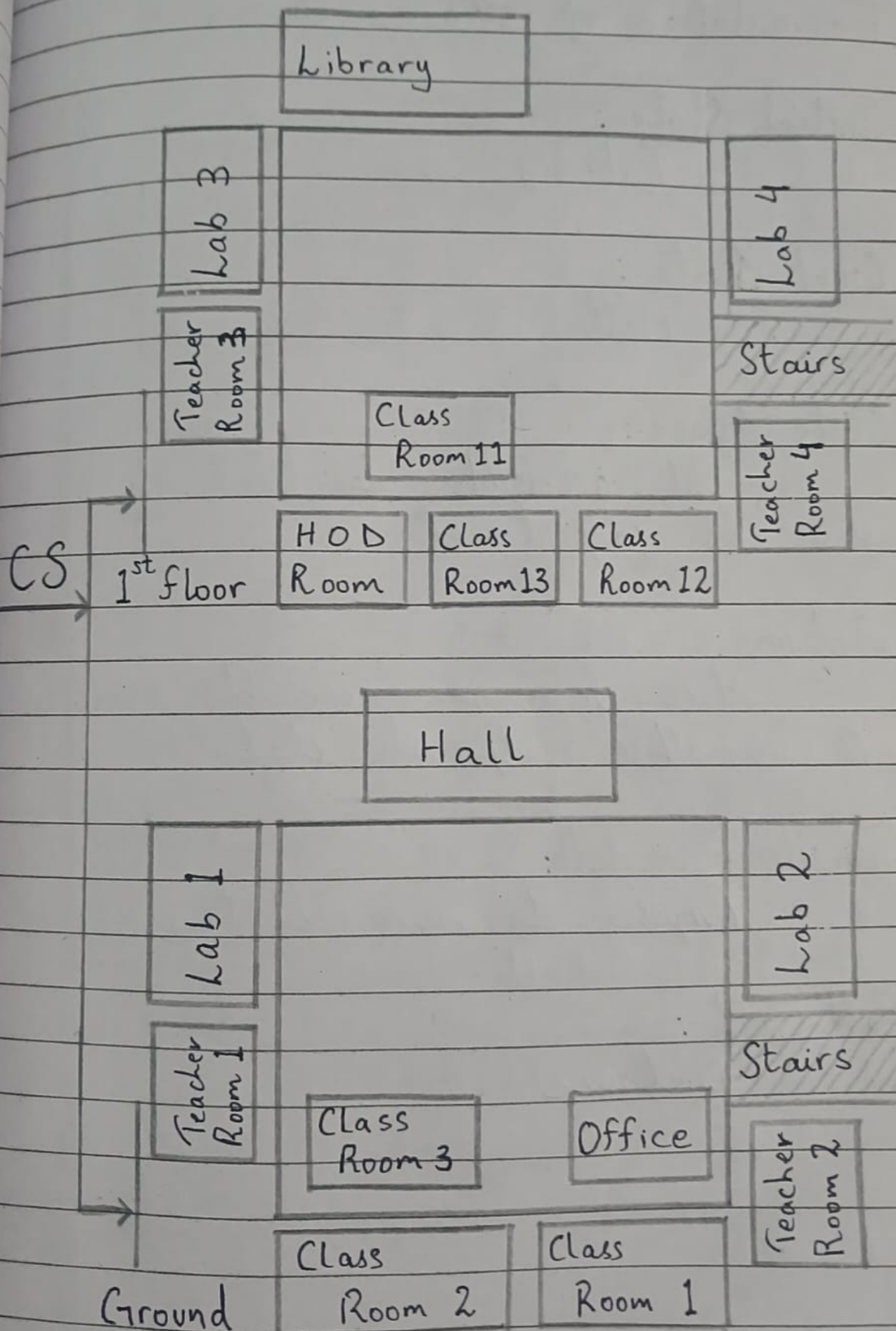


MAP OF CS

Date: 22/sep/2026



Formulation of the problem:

Initial State:

Library

Goal State:

Hall

State Space:

All accessible locations in
the CS department.

Action:

Move from one accessible state
to an other adjacent state

Transition model:

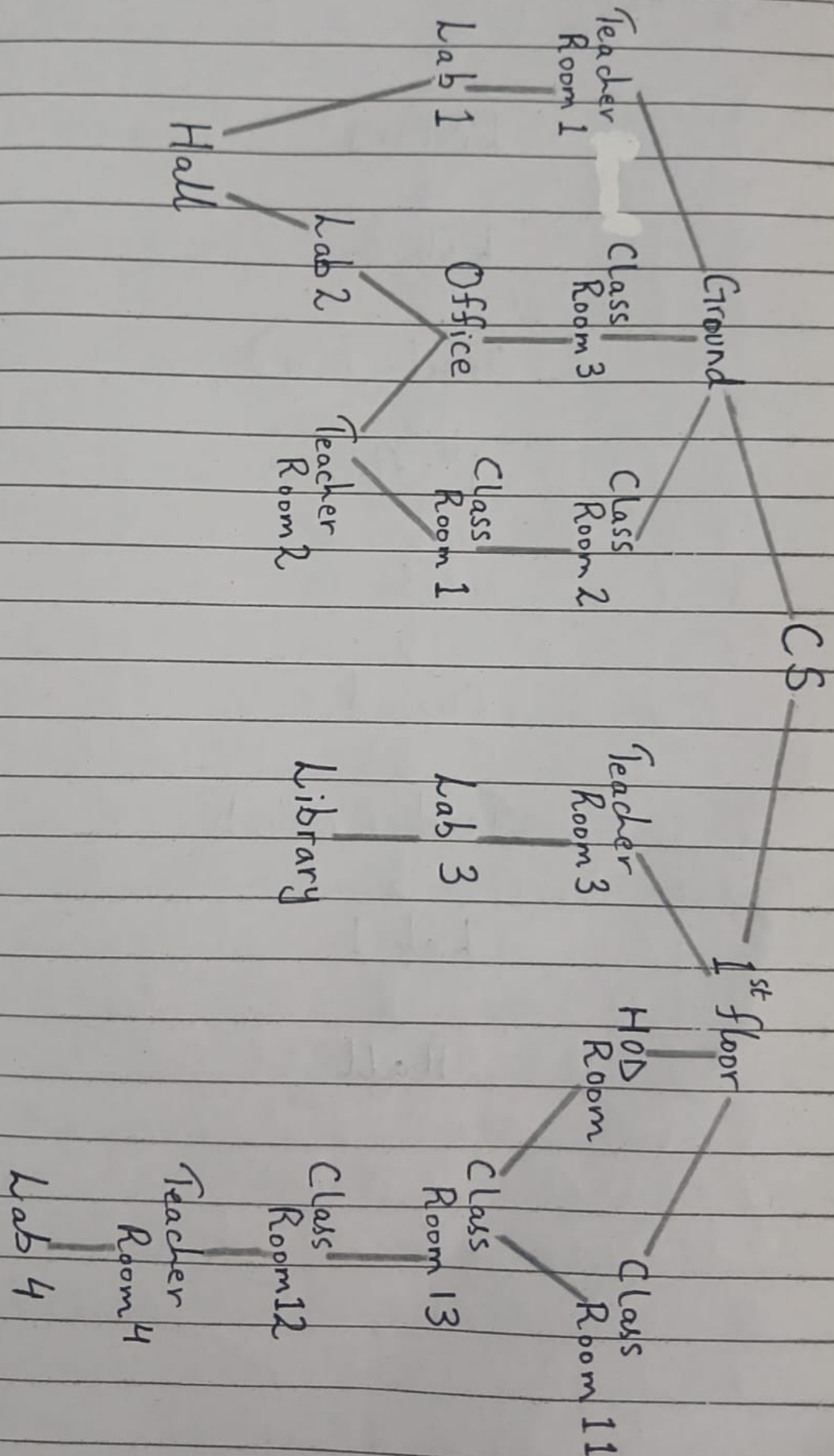
Purpose to move state.

Path cost:

Distance.

Cinaph

Date: _____



BFS Table

Date: _____

Initialization:

$Q = [\text{Library}]$

$V = [\text{Library}]$

deq Library & expand

$Q = [\text{Lab 3}]$

$V = [\text{Library, Lab 3}]$

deq Lab 3 & expand

$Q = [\text{Teacher room 3}]$

$V = [\text{Library, Lab 3, Teacher room 3}]$

deq Teacher room 3 & expand

$Q = [1^{\text{st}} \text{ floor}]$

$V = [,, , \text{Teacher room 3, } 1^{\text{st}} \text{ floor}]$

deq 1^{st} floor & expand

$Q = [\text{CS, HOD room, Class room 11}]$

$V = [,, , 1^{\text{st}} \text{ floor, CS, HOD room, Class room 11}]$

deq CS & expand

$Q = [\text{HOD room, class room 11, Ground}]$

$V = [,, , \text{Class room 11, Ground}]$

deq HOD room & expand

$Q = [\text{Class room 11, Ground, Class room 13}]$

$V = [,, , \text{Ground, Class room 13}]$

deq Class room 11 & expand

$Q = [\text{Ground, Class room 13,}]$

$V = [,, , \text{Ground, Class room 13}]$

deg Ground & expand

Q = [Class room 13, Teacher room 1, Classroom 3, Classroom 2]

V = [,, , Classroom 13, Teacher room 1, Classroom 3, Classroom 2]

deg Class room 13 & expand

Q = [Teacher room 1, Classroom 3, Class room 2, Classroom 13]

V = [,, , Class room 12]

deg Teacher room 1 & expand

Q = [Classroom 3, Class room 2, Classroom 12, Lab 1]

V = [,, , Lab 1]

deg Class room 3 & expand

Q = [Class room 2, Classroom 12, Lab 1, Office]

V = [,, , office]

deg Class room 2 & expand

Q = [Classroom 12, Lab 1, Office, Class room 1]

V = [,, , class room 1]

deg classroom 12 & expand

Q = [Lab 1, Office, Classroom 1, Teacher room 4]

V = [,, , Teacher room 4]

deg Lab 1 & expand

Q = [Office, Classroom 1, Teacher room 4, Hall]

V = [,, , Hall]

deg Office & expand

Q = [Classroom 1, Teacher room 4, Hall, Lab 2, Teacher room 2]

V = [,, , Teacher room 2]

deg Classroom 1 & expand

Q = [Teacher room 4, Hall, Lab 2, Teacher 2]

V = [] , Teacher room 2]

deg Teacher room 4 & expand

Q = [Hall, Lab 2, Teacher room 2, Lab 4]

V = [] , Lab 4]

deg Hall & expand

Q = [Lab 2, Teacher room 2, Lab 4]

V = [] , Lab 4]

deg Lab 2 & expand

Q = [Teacher room 2, Lab 4]

V = [] , Lab 4]

deg Teacher room 2 expand

Q = [Lab 4]

V = [] , Lab 4]

Shortest Path Library to Hall

Library



Lab 3



Teacher room 3



1st floor



CS



Ground



Teacher room 1



Lab 1



Hall